

Midterm (practice version - NOT GRADED), Math 170S, Spring 2019
Instructor: Elizaveta Rebrova

Printed name: _____

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SOLUTIONS

Instructions:

- Read problems very carefully. If you have any questions please ask.
- You should show all the details of your solution, the correct final answer alone gets a very small part of the credit. If you do not have time to write a proper solution, always attach your drafts and try to make them easier for us to follow.
- Your final answers do not need to be completely simplified (unless otherwise stated), e.g. a sum of several decimal fractions would be completely fine. Ask if in doubt.
- Calculators and books are not allowed. Phones, computers and any other communication devices are not allowed too.
- You can bring a letter size piece of paper with formulas and anything helpful for you (please be reasonable and do not squeeze out the whole textbook inside one page :)). Please bring your own writing utensils. We provide all necessary distribution tables and extra draft paper. If you need more draft paper please raise your hand.
- Please have your ID ready so we can check it without disturbing your work.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
Total:	50	

1. For this problem: provide short explanations only.

A candidate for a certain position is preparing for the elections. More than 50% positive votes will secure her win. After the preliminary polling, she computes confidence intervals for the fraction of people in her favor, with the confidence level $1 - \alpha$: a one-sided interval $[a, b]$ for the lower estimate; and a two-sided interval $[c, d]$.

(a) (2 points) What is b ?

$b = 1$ (maximal possible fraction of people supporting the candidate)

(b) (2 points) What can be bigger, a or c ?

$a \geq c$ ($P(p \in [a, b]) = P(p \in [c, d])$
 $b = 1 \geq d \Rightarrow a \geq c$)

(c) (2 points) What condition must be satisfied so that a candidate feels confident about her win? (with probability $1 - \alpha$)

$a \geq \frac{1}{2}$ (or $c \geq \frac{1}{2}$, depending on which confidence interval is used)

(d) (2 points) (Not about these elections any more :) $X_1, \dots, X_n$ is a sample from the distribution $Unif[0, \theta]$. Is the estimator $\tilde{\theta} = 2\bar{X}$ biased or not?

$$\bar{X} = \frac{\sum X_i}{n} \quad E\bar{X} = \frac{\sum E X_i}{n} = \frac{\sum_{i=1}^n \theta/2}{n} = \frac{\theta}{2}$$

$$E\tilde{\theta} = E2\bar{X} = \theta \Rightarrow \text{unbiased}$$

(e) (2 points) True or false: in the example above, if $X_{(n)}$ is a sufficient statistic, then $X_{(n)}^2$ is also sufficient?

True, since all $x_i \geq 0 \Rightarrow$ square is an invertible function (for $x \geq 0$, x^2 is invertible) $\Rightarrow$ u^2 is another sufficient statistic.

Hint: what are the possible values of θ ?

2. (10 points) You have a biased coin with θ being a probability of heads. You believe that θ is distributed ~~uniformly~~ according to the ~~uniform distribution~~

$$f_{\theta}(\theta) = 2 - 4|0.5 - \theta| \text{ for } 0 \leq \theta \leq 1$$

Find the MAP estimate of θ , if you observed exactly one heads in n experiments? (Get a general formula for any $n = 1, 2, \dots$)

if $\theta \leq 1/2$:

$$f_{\theta}(\theta) = 2 - 4(\frac{1}{2} - \theta) = 4\theta$$

$$f_{\theta}(\theta | x_1, \dots, x_n) = \frac{f_{\theta}(\theta) \cdot f(x_1, \dots, x_n | \theta)}{\int_{\theta'} f_{\theta}(\theta') \cdot f(x_1, \dots, x_n | \theta') d\theta'}$$

← This is more probable case (just one heads → heads are unlikely → θ is small) so we start with it.
"const" does not depend on θ

$$f(x_1, \dots, x_n | \theta) = \theta (1-\theta)^{n-1} \binom{n}{1}$$

$$f(\theta | x_1, \dots, x_n) = 4\theta \cdot \theta (1-\theta)^{n-1} \cdot \frac{1}{\text{const}}$$

$$f'(\theta | x_1, \dots, x_n) = 4 \cdot 2\theta (1-\theta)^{n-1} \cdot \frac{1}{\text{const}} - \frac{4\theta^2 (n-1)(1-\theta)^{n-2} \cdot \frac{1}{\text{const}}}{\text{const}} = 0$$

$$2(1-\theta) - \theta(n-1) = 0 \quad \theta = \frac{2}{n+1}$$

if $\frac{2}{n+1} \leq \frac{1}{2}$, namely,
otherwise

$$\boxed{\begin{matrix} n \geq 3, & \hat{\theta} = \frac{2}{n+1} \\ n < 3, & \hat{\theta} = \frac{1}{2} \end{matrix}}$$

Check your answer: what is the best θ you got

- for 5 experiments (coin flips)?

$$\frac{2}{6} = \frac{1}{3}$$

- for 2 experiments (coin flips)?

$$\frac{1}{2}$$

Note that both numbers must be between 0 and 1!

if $\theta > \frac{1}{2}$:

$$f_{\theta}(\theta) = 2 + 4\left(\frac{1}{2} - \theta\right) = 4 - 4\theta$$

$$f(\theta | x_1, \dots, x_n) = 4(1-\theta)^n \cdot \theta \cdot n \frac{1}{\text{const}}$$

$$f'(\theta | x_1, \dots, x_n) = \frac{4(1-\theta)^n \cdot n - 4n^2(1-\theta)^{n-1} \cdot \theta}{\text{const}} = 0$$

$$(1-\theta) - n\theta = 0$$

$$\theta(n+1) = 1$$

$$\theta = \frac{1}{n+1} \quad \text{— this is impossible for } n \geq 1 \text{ since } \theta > \frac{1}{2}$$

3. Consider a sequence of independent coin tosses, and let θ be the probability of heads at each toss.

(a) (5 points) Let k be a fixed number, and N is the number of tosses until k -th head occurs. Find maximum likelihood estimator of θ (depending of N).

$$MLE = \underset{\theta}{\operatorname{argmax}} \prod_{i=1}^n P(X_i = x_i), \text{ where } X_1, \dots, X_n - \text{experiments} \\ \text{and } x_1, \dots, x_n - \text{outcomes}$$

In our case - only one experiment was performed
(as only one number N was observed)

$$\# MLE = \underset{\theta}{\operatorname{argmax}} P(\text{number of tosses until } k\text{-th head} = N) \\ = \underset{\theta}{\operatorname{argmax}} \binom{N-1}{k-1} \theta^k (1-\theta)^{N-k} \quad \text{if } N \geq k \text{ / 0 otherwise} \\ \left\{ \text{exactly } k-1 \text{ successes in } N \text{ tosses +} \right. \\ \left. \text{last one is success} \right\}$$

$$\left[\binom{N-1}{k-1} \theta^k (1-\theta)^{N-k} \right]' = \\ = \text{const} \cdot [k \theta^{k-1} (1-\theta)^{N-k} - (N-k) \theta^k (1-\theta)^{N-k-1}] = 0 \\ k(1-\theta) - (N-k)\theta = 0 \Rightarrow \boxed{\hat{\theta} = k/N}$$

(b) (5 points) Now let n be a fixed number, and K is the number of heads in n tosses. Find maximum likelihood estimator of θ (depending on K).

$$\# MLE = P(\# \text{ successes in } n \text{ tosses} = K) = \\ = \binom{n}{K} \theta^K (1-\theta)^{n-K} \quad (\text{Binomial density})$$

$$\left[\binom{n}{K} \theta^K (1-\theta)^{n-K} \right]' = \text{const} [k \theta^{k-1} (1-\theta)^{n-k} - (n-k) \cdot \\ \cdot \theta^k \cdot (1-\theta)^{n-k-1}] = 0 \\ k(1-\theta) - (n-k)\theta = 0 \Rightarrow \boxed{\hat{\theta} = \frac{K}{n}}$$

4. (10 points) Let $X_1, X_2, \dots, X_n$ are samples taken from the uniform distribution

$$\text{Unif}([\theta, \theta + 1]).$$

What is the length of a confidence interval with confidence coefficient $1 - \alpha$ for the estimate $X_{(1)}$ for parameter θ ?

It makes sense to consider one-sided confidence interval, since $X_{(1)} \geq \theta$ always
 $(X_i \sim \text{Unif}[\theta, \theta + 1] \Rightarrow \text{all } X_i's \geq \theta)$

So, we need to find maximum length of $[X_{(1)}, X_{(1)} - c_\alpha]$, such that

$$\mathbb{P}(\theta \in [X_{(1)}, X_{(1)} - c_\alpha]) = 1 - \alpha \Rightarrow$$

$$1 - \alpha = \mathbb{P}\{X_{(1)} - c_\alpha \leq \theta\} = 1 - \mathbb{P}\{X_{(1)} - c_\alpha > \theta\}$$

$$= 1 - \mathbb{P}(X_1, \dots, X_n > c_\alpha + \theta) =$$

independence!

$$= 1 - \prod \mathbb{P}(X_i > c_\alpha + \theta) =$$

all samples have same distribution!
 $\stackrel{!}{=} 1 -$

$$\left(\mathbb{P}(X_1 > c_\alpha + \theta) \right)^n = 1 - \left(\mathbb{P}(X_1 \in [c_\alpha + \theta, \theta + 1]) \right)^n$$

$$= 1 - (1 - c_\alpha)^n$$

$$\text{So, } (1 - c_\alpha)^n = \alpha$$

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$$c_\alpha = 1 - \alpha^{1/n}$$

5. (10 points) Let $X_1, X_2, \dots, X_n$ be samples taken from some distribution with unknown density function. However, you know that $\mathbb{E}X = \alpha/\beta$ and $\text{Var } X = \alpha/\beta^2$, where α and β are two parameters of the density function (and there is no third parameter). What method could we use to find good estimates for α and β ? What are these estimates?

Solution:

Since we do not know density function, we cannot find maximum likelihood estimate, so let us find method of moments estimate. There are 2 parameters, so we need to find 2 moments.

First moment:

$$\mu_1 = \overline{X} = \tilde{\alpha}/\tilde{\beta}$$

Second moment:

$$\mu_2 = \frac{1}{n} \sum X_i^2 = \mathbb{E}(X^2) = (\mathbb{E}X)^2 + \text{Var } X = \tilde{\alpha}^2/\tilde{\beta}^2 + \tilde{\alpha}/\tilde{\beta}^2.$$

Solving system of equations for $\tilde{\alpha}$ and $\tilde{\beta}$, we get

$$\mu_2 - \mu_1^2 = \tilde{\alpha}/\tilde{\beta}^2$$

$$\frac{\mu_1^2}{\mu_2 - \mu_1^2} = \tilde{\alpha}$$

$$\frac{\mu_1^2}{\mu_2 - \mu_1^2} \frac{1}{\mu_1} = \frac{\mu_1}{\mu_2 - \mu_1^2} = \tilde{\beta}$$

Answer:

$$\tilde{\alpha} = \frac{\overline{X}^2}{\frac{1}{n} \sum X_i^2 - \overline{X}^2};$$

$$\tilde{\beta} = \frac{\overline{X}}{\frac{1}{n} \sum X_i^2 - \overline{X}^2},$$

where

$$\overline{X} = \frac{1}{n} \sum X_i.$$